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ORIGINAL PAPER

Enhancing crop disease classification and severity assessment through Fuzzy Rank-based ensemble learning approach

Yassamina Medjadba[1] · Hamza Drid[1] · XianChuan Yu[2]

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Abstract

Plant diseases affect crop yields, making accurate detection and severity assessment vital. This study proposes a framework combining pre-trained CNNs with a fuzzy rank-based ensemble for improved classification. Three CNNs are integrated, with predictions guided by confidence assessments. For diseased images, the model localizes affected areas and uses the Analytic Hierarchy Process (AHP) to assess severity as mild, moderate, or severe. Results show the method outperforms existing approaches in both detection and severity estimation.

Keywords Transfer learning · Fuzzy rank-based ensemble · AHP · Score-based severity prediction

1 Introduction

The agricultural sector plays a vital role in ensuring food security and supporting economic growth [1]. The ongoing growth in the global population highlights the urgent demand for resilient and sustainable farming practices. However, plant diseases present a major challenge to agriculture by reducing crop yields, lowering quality, and disrupting ecosystem balance. Conventional detection methods rely on labor-intensive expert observation, often lacking accuracy and accessibility, especially in rural areas. Climate change further exacerbates disease outbreaks [2].

Recent advances in deep learning and computer vision have transformed disease detection through automation and accuracy. Transfer learning improves performance in data-scarce settings by reusing knowledge from existing models.

Despite progress, the need for reliable models remains. This study proposes a fuzzy rank-based ensemble approach to enhance plant disease detection and severity estimation. It combines three lightweight transfer learning models: MobileNetV3, SE-EfficientNetB0, and SE-Xception, using a fuzzy ranking mechanism for final prediction. Additionally, a score-based severity assessment, inspired by the Analytic Hierarchy Process (AHP), categorizes infections as mild, moderate, or severe.

The key contributions of this research are summarized below:

- A framework combining three transfer learning models with a fuzzy rank-based ensemble for disease classification and infection localization.
- The ensemble calculates fuzzy ranks using nonlinear functions on model scores; predictions are made by selecting the class with the lowest combined rank.
- A severity assessment method, inspired by AHP, categorizes infected images as mild, moderate, or severe.

The paper is organized as follows: Section 2 reviews related work, Section 3 outlines the methodology, Section 4 presents experiments, and Section 5 highlights conclusion and future work.

2 Related work

In recent years, image-based methods using machine learning models, such as SVM, Random Forest, and ANN, have

✉ Yassamina Medjadba
y.medjadba@univ-batna2.dz

Hamza Drid
h.drid@univ-batna2.dz

XianChuan Yu
yuxanchuan@163.com

1 Department of Computer Science, University of Batna 2, Batna, Algeria

2 School of Artificial Intelligence, Beijing Normal University, Beijing, China

Fig. 1 Different samples from PlantDoc dataset

leveraged features such as shape, color, and texture for plant disease detection. However, their high computational cost limits efficiency.

Deep learning has revolutionized plant disease classification, driving noteworthy innovations. Ullah et al. [3] developed DTomatoDNet that reached 99.34% accuracy for tomato leaf diseases. Batchuluun et al. [4] combined PI-CNN and PI-GAN, achieving accuracies of 97.27% and 80.36% on PlantVillage and PlantDoc, respectively. DeepPlantNet [5] reported 98.49% accuracy for eight diseases and 99.85% for three. A trilinear CNN with bilinear pooling [6] achieved 99.7% accuracy in controlled conditions and 75.58% in real-world conditions. Li et al. [7] introduced PMVT, a MobileViT-based model, achieving 93.6% accuracy for wheat, 85.4% for coffee, and 93.1% for rice.

Ensemble models excel in disease classification, with Hatice et al. [8] achieving 99.72% accuracy for wheat using DenseNet201, InceptionV3, and other algorithms, though requiring more labeled data. Zasim et al. [9] reached 99.1% accuracy on rice leaf diseases with the E2ETCA framework, but its performance on larger datasets is unclear.

Plant disease classification faces challenges due to species diversity, overlapping symptoms, and complex models.

3 Material and methods

3.1 Dataset collection

Leaf disease detection begins with collecting and preprocessing images, resized to 224×224 and normalized. The PlantDoc dataset, captured in real-world conditions, is used to evaluate the model's practical effectiveness (Figure 1).

The PlantDoc dataset contains around 2,598 images across 27 classes and 13 species. Originally sourced online (29,000 images), it was filtered to remove non-leaf, duplicate, and lab images, with labels manually verified and corrected.

The dataset is divided into two subsets: 80% of the data is assigned for training, while the remaining 20% is reserved for validation.

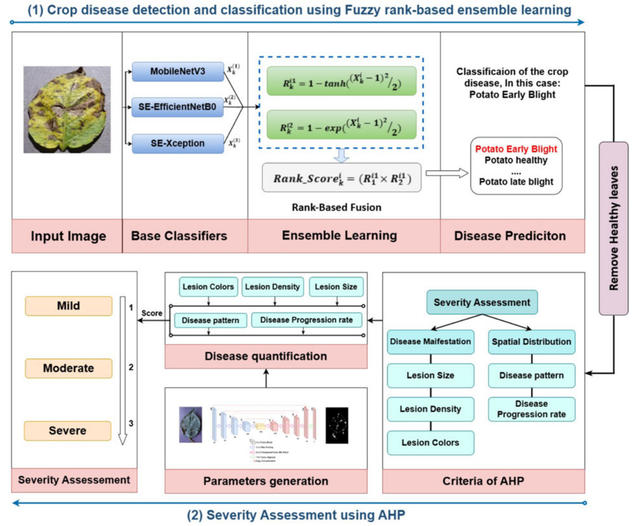

Fig. 2 Architecture of the proposed method

3.2 Proposed method

This section presents the proposed architecture, which addresses two main tasks: disease classification and infection severity assessment.

For classification, an ensemble of MobileNetV3, SE-EfficientNetB0, and SE-Xception is combined using a fuzzy rank-based method. For severity assessment, the Analytic Hierarchy Process (AHP) evaluates infection levels based on classification outputs. Figure 2 illustrates the overall framework.

3.2.1 MobileNetV3 transfer learning

MobileNetV3 is a lightweight architecture optimized for efficiency using the NetAdapt algorithm, designed for specific hardware platforms [10], and features the h-swish activation function.

$$\mathrm{h_{swish}}(x) = x \cdot \gamma(x) \tag{1}$$

$$\gamma(\mathrm{x}) = ReLU6(x + 3)/6 \tag{2}$$

The model uses depthwise separable convolutions, splitting standard convolution into depthwise and pointwise operations to reduce size and computation.

MobileNetV3 has two variants: large and small. The large version is chosen for this study due to its deeper architecture and higher classification accuracy. It includes 2D convolutions, bottleneck layers at multiple resolutions, SE layers for channel recalibration, and a dense layer with 1024 units for feature extraction.

For transfer learning, the pre-trained MobileNetV3 (ImageNet) is adapted by replacing its fully connected layers with a Global Average Pooling layer and a softmax classification layer, improving generalization to the target dataset.

3.2.2 SE-EfficientNetB0 transfer learning

EfficientNet [11] is a CNN optimized for efficiency, reducing FLOPS while maintaining accuracy through compound scaling. The architecture consists of a stem, seven blocks, and a final layer. EfficientNetB0 has 237 layers and 5.3M parameters, while EfficientNetB7 has 813 layers and 66M parameters, both utilizing MBConv layers.

EfficientNetB0 is chosen for its balance of accuracy and efficiency. Using ImageNet pre-trained weights, the backbone is frozen, and a new classification head is added, consisting of a Global Average Pooling layer, a Dense layer, and a softmax output. An SE block [12] is also integrated to enhance channel-wise feature recalibration and improve performance with minimal overhead.

3.2.3 SE-Xception transfer learning

Xception, an extension of Inception, features 36 convolutional layers and a residual design [13]. It uses depthwise separable convolutions and 1×1 pointwise convolutions to efficiently capture spatial and cross-channel correlations, achieving high accuracy with fewer parameters.

For transfer learning, the pre-trained Xception model is used, with its convolutional layers frozen to retain learned features. A new classification head is added, consisting of an SE block [12] for adaptive feature recalibration, a Global Average Pooling layer, and a Dense layer with softmax activation. This configuration enhances learning efficiency and generalization to the target dataset.

3.2.4 Proposed ensemble approach for final classification

This section outlines the fuzzy rank-based ensemble method, which integrates predictions from three pre-trained models: CNN_1 (*MobileNetV3*), CNN_2 (*SE-EfficientNetB0*), and CNN_3 (*SE-Xception*). The approach enhances classification accuracy by prioritizing confident predictions and reducing uncertainty through fuzzy rankings.

Each model generates confidence scores $X_k^{(m)}$ for class k. These scores are transformed into fuzzy ranks using two nonlinear functions:

$$R_k^{(1)} = 1 - \tanh\left(\frac{(X_k^{(m)} - 1)^2}{2}\right), \quad (3)$$

$$R_k^{(2)} = 1 - \exp\left(\frac{(X_k^{(m)} - 1)^2}{2}\right). \quad (4)$$

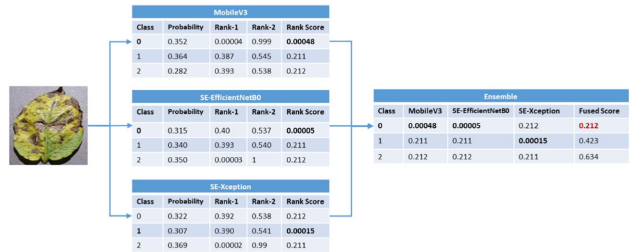

Fig. 3 Example of the ensemble model process with a 3-class dataset image

The *Rank Score* for each class is computed by combining both fuzzy ranks:

$$\text{Rank\_Score}_k^{(i)} = R_k^{(1)} \times R_k^{(2)}, \quad (5)$$

The *Fused Score* for each class is the sum of the Rank Scores from all models:

$$FS_k = \sum_{i=1}^{3} \text{Rank\_Score}_k^{(i)}, \quad (6)$$

The final predicted class $\hat{k}$ is the one with the minimum *Fused Score*:

$$\hat{k} = \arg\min_k FS_k. \quad (7)$$

This method refines confidence scores and ensures that only reliable predictions influence the final classification decision.

This selection criterion ensures that the class with the highest consensus confidence (indicated by the lowest Fused Score) is chosen as the final predicted class (Figure 3).

3.2.5 Complexity Analysis of the Proposed Method

This section analyzes the computational complexity of each step in the ensemble method.

Each pre-trained model (CNN_1, CNN_2, CNN_3) processes an input image of size $w \times h \times d$ to generate confidence scores for c classes. Let P_{CNN_i} be the number of parameters in model i and F_{CNN_i} be the floating-point operations (FLOPs) per forward pass.

The total complexity of classification across all three models is:

$$\mathcal{O}\left(\sum_{i=1}^{3} F_{CNN_i}\right) \quad (8)$$

The total number of parameters across all models is:

$$\mathcal{O}\left(\sum_{i=1}^{3} P_{CNN_i}\right) \tag{9}$$

For the fuzzy ranking step, each model's output is transformed using nonlinear functions (constant complexity per class), yielding:

$$\mathcal{O}(m \cdot c) \tag{10}$$

where $m = 3$ is constant.

Finally, the complexity of class selection (finding the class with the minimum fused score) is:

$$\mathcal{O}(c) \tag{11}$$

Thus, the overall complexity is:

$$\mathcal{O}\left(\sum_{i=1}^{3} F_{CNN_i} + \sum_{i=1}^{3} P_{CNN_i} + c\right) \tag{12}$$

Where$\sum_{i=1}^{3} F_{CNN_i}$ corresponds to CNN inference, $\sum_{i=1}^{3} P_{CNN_i}$ accounts for model parameters, and c represents fuzzy ranking and class selection.

Since c is small, the main computational cost comes from CNN inference and model parameters, with minimal impact from the fuzzy ranking step.

3.2.6 Measuring the Severity of Disease Classification

Accurate plant disease severity assessment is key for effective management. This study proposes a score-based approach using the AHP for better severity evaluation and decision-making. The AHP [14] combines qualitative and quantitative analysis to solve complex, conflicting problems.

Criteria Selection

In the first phase, disease severity criteria are selected based on their impact, measurability, and relevance to agronomic decisions. With expert input, the criteria are divided into two categories (Figure 1):

1. **Disease Manifestation:** Includes three sub-factors:
   - **Lesion Size (S):** Extracted via image segmentation.
   - **Lesion Density (D):** Calculated as the ratio of affected pixels to total leaf area.
   - **Lesion Color (C):** Evaluated using heatmaps as an indicator of disease progression [15].

   We employ a **UNet** model for disease segmentation.
2. **Spatial Distribution:** Evaluates lesion spread, includes:
   - **Disease Pattern (P) :** Whether lesions exhibit a localized or systemic distribution.
   - **Disease Progression Rate (PR):** Analyzed through lesion dispersion within and between leaves.

Table 1 presents the severity assessment criteria, where all factors are considered equally important, without assigned weights.

Decision-Making

In the second setup, disease manifestation and spatial distribution criteria are first calculated from the input image. Disease manifestation is evaluated based on the size, density, and color of the lesions, resulting in the following score :

$$S_1 = \frac{\text{(S)-score} + \text{(D)-score} + \text{(C)-score}}{3} \tag{13}$$

where 3 represents the total number of criteria. Similarly, spatial distribution is evaluated based on the distribution pattern and disease progression, resulting in the second score, S_2.

$$S_2 = \frac{\text{(P)-score} + \text{(PR)-score}}{2} \tag{14}$$

The overall Severity Score is determined by combining both criteria S_1, S_2, and is evaluated as follows:

Score 1: Mild disease, **Score 2**: Moderate disease, **Score 3**: Severe disease.

This approach integrates both manifestation and spatial distribution, offering a comprehensive view of disease severity. As a result, it enables more accurate disease management.

4 Experimental results and discussion

This section includes details about the experimental settings, metrics, visualization and comparison results.

4.1 Experimental settings

The proposed approach is implemented in Python and accelerated using GPU. The hardware setup includes an Intel® Core^{TM} i7-10750H CPU (2.60 GHz) and NVIDIA GeForce RTX 2060 for both training and testing. The training hyperparameters are provided in Table 2.

Table 1 Severity assessment criteria

Category	Criteria	Severity Intervals	Score
Disease Manifestation	Lesion Size	Lesions < 0.2	1
		$0.2 \leq$ lesions < 0.5	2
		Lesions ≥ 0.5	3
	Lesion Density	Density $< 20\%$	1
		$20\% \leq$ density $< 30\%$	2
		Density $\geq 30\%$	3
	Lesion Color	Yellow	1
		Orange	2
		Red	3
Spatial Distribution	Disease Pattern	Localized	1-2
		Systemic	2-3
	Progression Rate	Slow	1
		Moderate	2
		Rapid	3

Table 2 Training hyperparameters for the proposed model

Hyperparameter	Value
Batch size	32
Epochs	50
Momentum	0.9
Learning rate	0.01–0.00001
Optimizer	Adam
Loss function	Sparse Categorical Crossentropy
Dropout rate	0.3

4.2 Metrics used for evaluation

The model is evaluated using Accuracy, Precision, Sensitivity, and F1 Score. The definitions are as follows:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP+TN+FP+FN}} \tag{15}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{16}$$

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{17}$$

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}} \tag{18}$$

4.3 Comparative evaluation of plant disease detection

In this study, we used MobileNetV3, SE-EfficientNetB0, and SE-Xception as base models with pre-trained ImageNet weights. The final layer is fine-tuned on our dataset, and models are trained for 50 epochs with a batch size of 32 using Sparse Categorical Crossentropy, Adam optimizer, and dropout. Data augmentation techniques are applied, and model predictions are combined using a fuzzy rank-based ensemble. Performance was evaluated using accuracy, F1-score, and other metrics on a validation set.

For the selected dataset, the classification results from different models, followed by the fuzzy rank-based ensemble model, are presented in Table 3.

In this experiment, SE-EfficientNetB0 exhibits the best performance, achieving an accuracy of 99.68%. MobileNetV3 follows with an accuracy of 98.72%, whereas SE-Xception, VGG19, and MobileNetV2 achieve accuracies of 97.30%, 97.78%, and 97.49%, respectively. In contrast, Inception shows comparatively lower accuracy at 87.13%. The proposed ensemble method achieves an accuracy of 99.61% and an F1 Score of 99.70%. Sensitivity, a critical metric for plant disease classification, is notably high, reaching 99.66% for PlantDoc dataset.

We also conducted an ablation study to evaluate the contribution of individual components in our ensemble method as shown in Table 4.

We assessed the performance of SE-EfficientNetB0, MobileNetV3, and SE-Xception. Removing the SE module from EfficientNetB0 and Xception decreased performance, emphasizing the importance of channel attention. Our fuzzy-ranked ensemble outperformed majority voting, offering better accuracy and robustness. These results highlight the benefits of combining SE modules with fuzzy ranking for improved classification and efficiency.

The training vs. validation loss graphs (Figure 4) show that none of the pre-trained models exhibited significant overfitting, as the validation loss closely matched the training loss in the final epochs.

Table 3 Performance metrics for different models on PlantDoc datasets

Dataset	Model	Precision (%)	Sensitivity (%)	F1 Score(%)	Accuracy (%)
PlantDoc	**SE-EfficientNetB0**	**99.68**	**99.43**	**99.52**	**99.22**
	MobileNetV3	**98.72**	**98.83**	**98.68**	**98.61**
	SE-Xception	97.30	97.63	**97.25**	**97.27**
	Inception	87.13	87.40	87.24	87.44
	VGG19	97.78	97.75	97.60	97.27
	MobileNetV2	97.49	98.10	98.08	98.05
	Proposed Model	**99.75**	**99.66**	**99.70**	**99.61**

Table 4 Performance comparison of different Experiments on PlantDoc dataset

Experiment	Description	PlantDoc Accuracy (%)
EfficientNetB0	Base model	97.02
SE-EfficientNetB0	Base model	99.22
MobileNetV3	Base model	98.61
Xception	Base model	96.52
SE-Xception	Base model	97.27
Majority Voting Ensemble	Three base models with majority voting	99.55
Proposed Model	Fuzzy ranking-based ensemble	99.61

Table 5 The performance comparison of the proposed model against previous research for the PlantDoc dataset

Method	dataset	Accuracy (%)	Precision (%)	Sensitivity (%)	F1 Score(%)
Trilinear-CNN([6])	Plantdoc	75.58	-	-	74.44
PI-CNN,PI-GAN([4])	Plantdoc	80.36	67.00	67.87	66.13
MobileNetV3 ([16])	Plantdoc	83	-	-	83
RAFA-Net([17])	Plantdoc	85.54	86	86	86
Conv encoder-decoder([18])	Plantdoc	84.56	-	-	-
Proposed Method	Plantdoc	99.61	99.75	99.66	99.70

We also compared the proposed method with state-of-the-art techniques on the PlantDoc dataset. The performance results are shown in Table 5.

The PlantDoc dataset poses challenges due to real-world variations like lighting and background differences. As shown in Table 5, existing methods, such as Trilinear-CNN, PI-CNN, PI-GAN, MobileNetV3, Conv encoder-decoder, and RAFA-Net, achieve accuracies between 75.58% and 85.54%. However, PI-CNN and PI-GAN show 80.36% accuracy with lower Precision (67.00%), Sensitivity (67.87%), and F1 Score (66.13%). Many methods also lack detailed metrics, limiting robustness evaluation. In contrast, the proposed method achieves 99.61% accuracy with high Precision (99.75%), Sensitivity (99.66%), and F1 Score (99.70%), demonstrating its effectiveness for real-world agricultural applications.

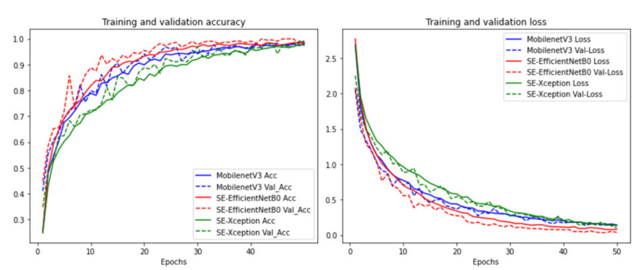

Fig. 4 Training and validation performance on PlantDoc datasets

4.4 Visualization and disease severity assessment

Once an image is classified as diseased, it undergoes segmentation to identify the affected regions, as shown in Figure 5. The segmented information is then used to establish specific criteria for evaluating disease severity.

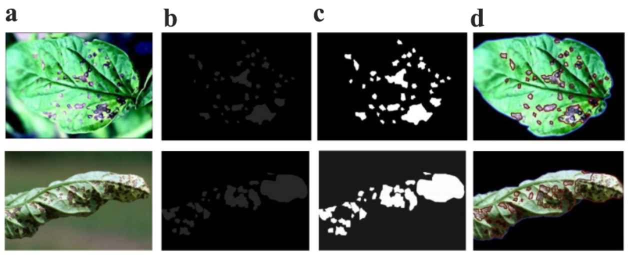

Fig. 5 Visualizing predictions of plant disease segmentation, (a) Input Image, (b) Ground truth, (c) Predicted mask, and (d) Segmented image, using UNet model

Fig. 6 Severity assessment results of plant disease samples

Figure 6 shows the severity assessment results, displaying disease conditions and severity levels with corresponding scores to assist agricultural experts in treatment planning.

First Image (Left): 0.53 cm lesions, 7% density, mild disease (Score1: 1.6, Score2: 1.5). Recommend mild pesticides and pruning.

Second Image (right): 0.6 cm lesions, 25% density, moderate severity (Score1: 2.3, Score2: 2). Recommend Pruning and observation.

This method aids in precise disease assessment and timely decision-making for crop sustainability.

5 Conclusion and future work

This paper presents a framework for plant disease identification that integrates image classification, and severity assessment. Using an ensemble of three pre-trained models, it improves classification accuracy through confidence-based predictions. The Analytical Hierarchy Process (AHP) aids treatment decisions by assessing disease progression. While the framework performs well, limitations include the exclusion of some diseases and a limited severity evaluation criteria. Future work will expand disease coverage, enhance model efficiency through quantization or distillation, and develop mobile applications to support real-world use and help farmers monitor crops more effectively.

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Author Contributions Yassamina Medjadba and Hamza Drid contributed substantially to all aspects of the work, including writing the main manuscript text. Yuxianchuan contributed to specific sections of the manuscript. All authors reviewed the final version of the manuscript.

Data Availability The dataset used in this study are publicly available in the following repository: **PlantDoc Dataset**: Accessible on Kaggle at the following URL: https://www.kaggle.com/datasets/nirmalsankalana/plantdoc-dataset.

Declarations

Conflicts of Interest The authors declare that they have no conflict of interest.

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